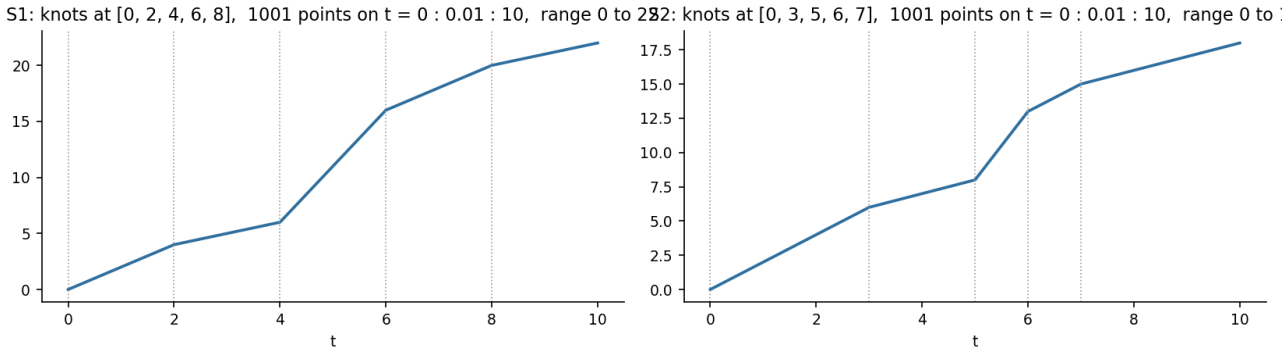
KAN approximation of S1 and S2: complete results

Experiments A (single edge, linear splines), B (single edge, cubic splines) and C (one hidden layer). Every fitted configuration is shown with its full curve, its error at every point, and its summary numbers. Data and generator from Nadia's ExampleForORF.m.

Setup

Item	Value
Targets	S1: knots [0,2,4,6,8]; S2: knots [0,3,5,6,7]; slopes [2,-1,4,-3,-1] for both; S(t) = sum_j A_j * max(0, t - theta_j)
Sample	1001 points, t = 0, 0.01, ..., 10 (files S1.txt, S2.txt; regenerated from the script, match to 4e-15)
KAN library	PyKAN 0.2.8 (PyTorch 2.2.2), grid_range [0, 10], grid updating off, seed 0
Training	L-BFGS, 200 steps, full batch, target scaled to [0,1] for optimisation and rescaled after
Errors	computed on all 1001 points in original units; worst error = max |KAN - S|; RMSE = root mean square; mean abs = mean |KAN - S
Reproduce	python dump_fits.py (writes all_fits.json); python make_result_figs.py; python build_results_pdf.py



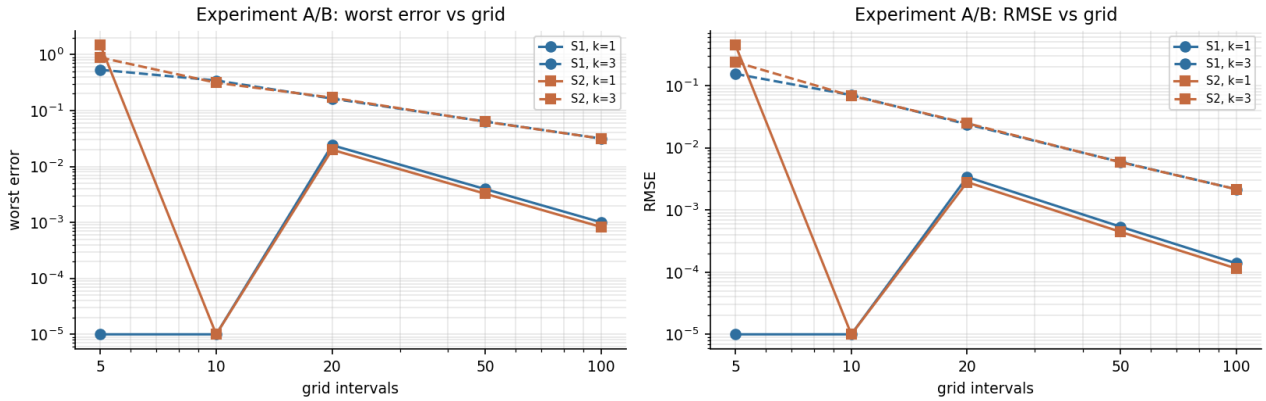
The two target functions. Dotted lines mark the knots.

1. All results in one table

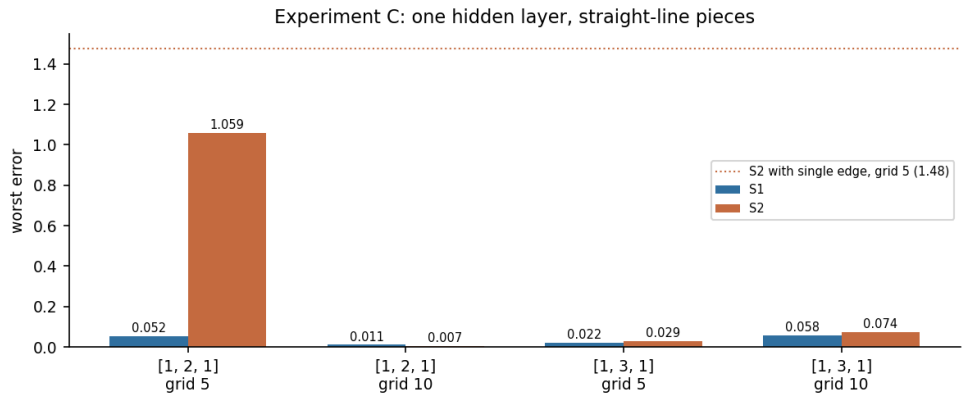
Sorted by experiment, then grid. 'Worst at t' is where the largest error occurs; for S2 on coarse grids it sits on a knot the grid cannot reach.

Exp	KAN width	Grid	Spline	Params	Fn	Worst error	RMSE	Mean abs	Worst at t
A	[1, 1]	5	linear k=1	12	S1	0 (exact)	0 (exact)	0 (exact)	7.27
A	[1, 1]	5	linear k=1	12	S2	1.4761	0.4475	0.3354	5.00
A	[1, 1]	10	linear k=1	17	S1	0 (exact)	0 (exact)	0 (exact)	6.22
A	[1, 1]	10	linear k=1	17	S2	0 (exact)	0 (exact)	0 (exact)	8.81
A	[1, 1]	20	linear k=1	27	S1	0.0240	0.0034	0.0014	0.00
A	[1, 1]	20	linear k=1	27	S2	0.0199	0.0028	0.0011	0.00
A	[1, 1]	50	linear k=1	57	S1	0.0039	0.0005	0.0002	0.20
A	[1, 1]	50	linear k=1	57	S2	0.0033	0.0004	0.0002	0.20
A	[1, 1]	100	linear k=1	107	S1	0.0010	0.0001	0 (exact)	0.10
A	[1, 1]	100	linear k=1	107	S2	0.0008	0.0001	0 (exact)	0.10
B	[1, 1]	5	cubic k=3	14	S1	0.5349	0.1542	0.1218	4.00
B	[1, 1]	5	cubic k=3	14	S2	0.8845	0.2394	0.1694	5.00
B	[1, 1]	10	cubic k=3	19	S1	0.3442	0.0703	0.0452	4.00
B	[1, 1]	10	cubic k=3	19	S2	0.3144	0.0684	0.0439	5.00
B	[1, 1]	20	cubic k=3	29	S1	0.1634	0.0238	0.0145	4.00
B	[1, 1]	20	cubic k=3	29	S2	0.1699	0.0248	0.0127	5.00
B	[1, 1]	50	cubic k=3	59	S1	0.0637	0.0059	0.0027	4.00
B	[1, 1]	50	cubic k=3	59	S2	0.0631	0.0059	0.0025	5.00
B	[1, 1]	100	cubic k=3	109	S1	0.0316	0.0022	0.0007	4.00
B	[1, 1]	100	cubic k=3	109	S2	0.0316	0.0022	0.0007	5.00
C	[1, 2, 1]	5	linear k=1	48	S1	0.0519	0.0183	0.0144	2.00
C	[1, 2, 1]	5	linear k=1	48	S2	1.0590	0.2815	0.1713	5.00
C	[1, 2, 1]	10	linear k=1	68	S1	0.0109	0.0019	0.0012	2.00
C	[1, 2, 1]	10	linear k=1	68	S2	0.0067	0.0018	0.0013	5.38
C	[1, 3, 1]	5	linear k=1	72	S1	0.0220	0.0034	0.0023	0.32
C	[1, 3, 1]	5	linear k=1	72	S2	0.0294	0.0052	0.0033	0.00
C	[1, 3, 1]	10	linear k=1	102	S1	0.0584	0.0128	0.0086	5.00
C	[1, 3, 1]	10	linear k=1	102	S2	0.0740	0.0168	0.0108	5.00

Summary charts



Experiments A and B: worst error (left) and RMSE (right) against grid size, log scales. Solid = linear splines, dashed = cubic.



Experiment C: worst error by configuration. Dotted line: the single-edge grid-5 result on S2, for comparison.

2. S1 versus S2, side by side

Experiment A: single edge [1,1], linear splines (k=1)

KAN width	Grid	Params	S1 worst	S1 RMSE	S2 worst	S2 RMSE	S2/S1 worst ratio
[1, 1]	5	12	0 (exact)	0 (exact)	1.4761	0.4475	n/a
[1, 1]	10	17	0 (exact)	0 (exact)	0 (exact)	0 (exact)	n/a
[1, 1]	20	27	0.0240	0.0034	0.0199	0.0028	0.8x
[1, 1]	50	57	0.0039	0.0005	0.0033	0.0004	0.8x
[1, 1]	100	107	0.0010	0.0001	0.0008	0.0001	0.8x

Experiment B: single edge [1,1], cubic splines (k=3)

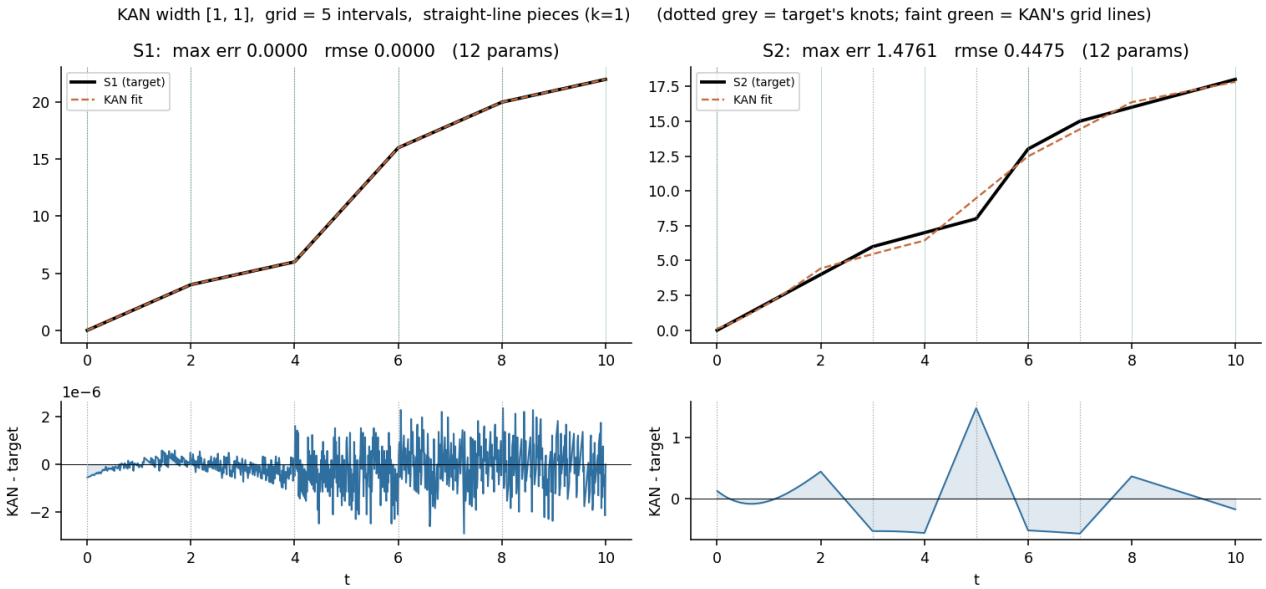
KAN width	Grid	Params	S1 worst	S1 RMSE	S2 worst	S2 RMSE	S2/S1 worst ratio
[1, 1]	5	14	0.5349	0.1542	0.8845	0.2394	1.7x
[1, 1]	10	19	0.3442	0.0703	0.3144	0.0684	0.9x
[1, 1]	20	29	0.1634	0.0238	0.1699	0.0248	1.0x
[1, 1]	50	59	0.0637	0.0059	0.0631	0.0059	1.0x
[1, 1]	100	109	0.0316	0.0022	0.0316	0.0022	1.0x

Experiment C: one hidden layer, linear splines (k=1)

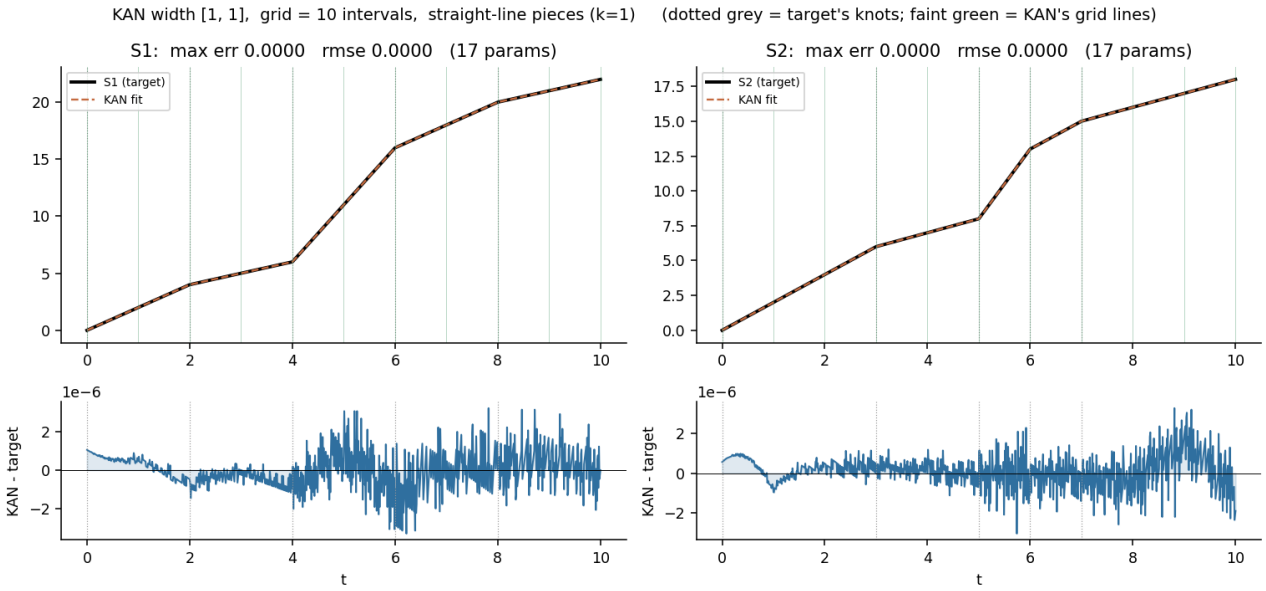
KAN width	Grid	Params	S1 worst	S1 RMSE	S2 worst	S2 RMSE	S2/S1 worst ratio
[1, 2, 1]	5	48	0.0519	0.0183	1.0590	0.2815	20.4x
[1, 2, 1]	10	68	0.0109	0.0019	0.0067	0.0018	0.6x
[1, 3, 1]	5	72	0.0220	0.0034	0.0294	0.0052	1.3x
[1, 3, 1]	10	102	0.0584	0.0128	0.0740	0.0168	1.3x

3. Every fit, one configuration per figure

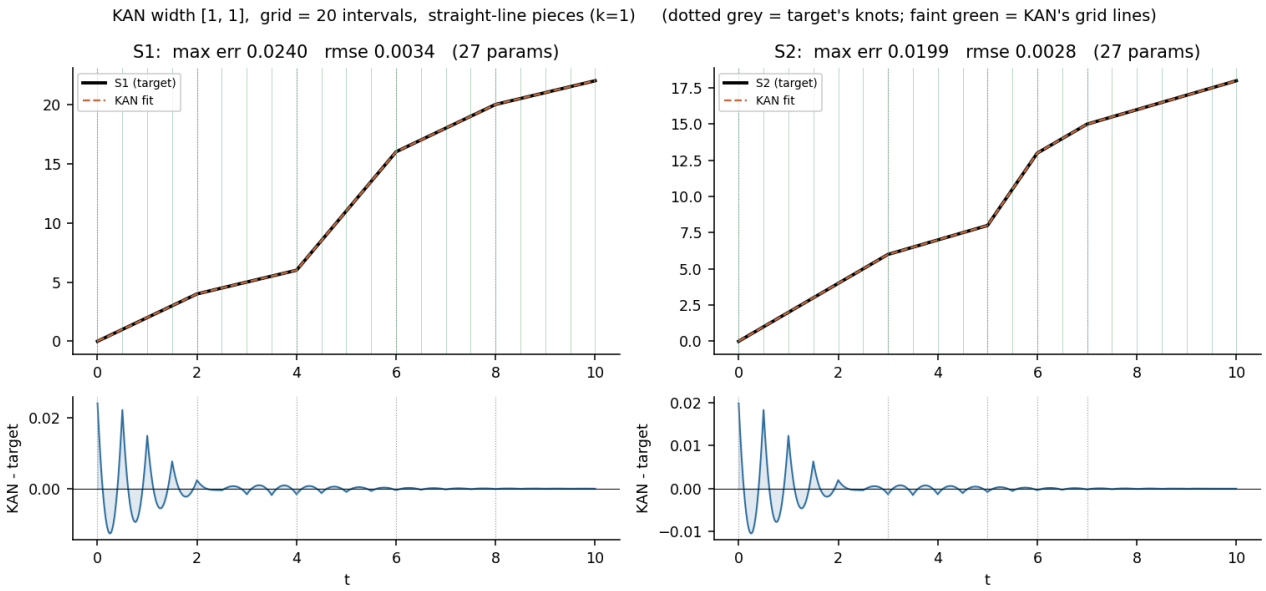
Top row: target (black) and KAN fit (orange dashed), with the target's knots (dotted grey) and the KAN's grid lines (faint green). Bottom row: the error KAN - target at every point. Note the vertical scale of the error panels differs between figures; exact fits show noise of order 1e-6.



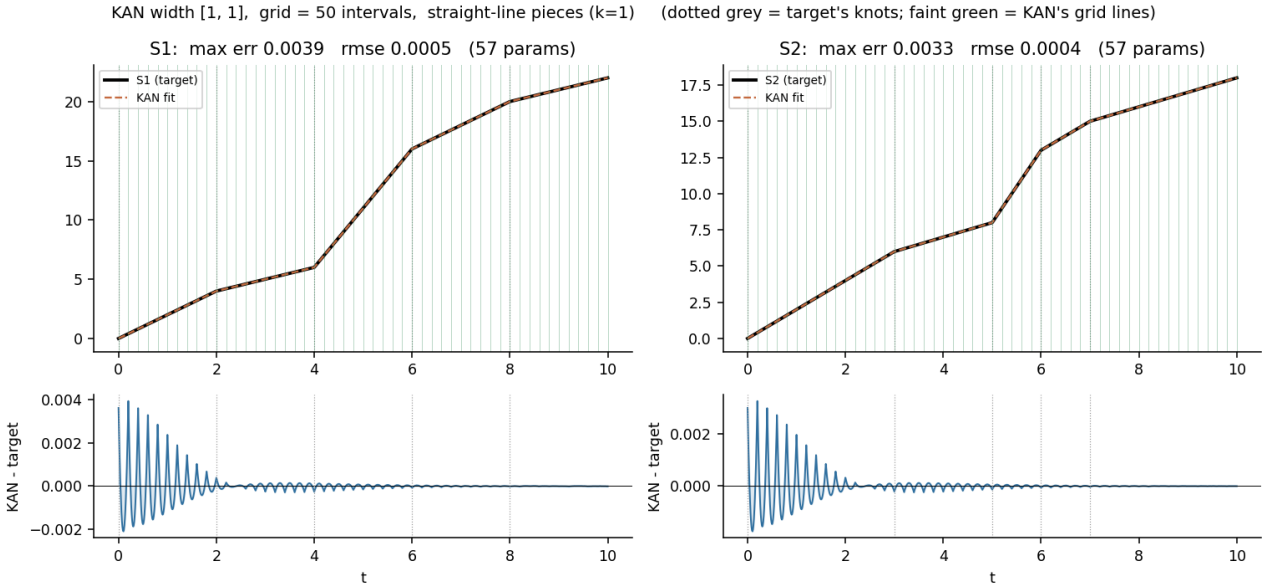
Experiment A. Width [1, 1], grid 5, k=1, 12 parameters. S1: worst 0 (exact), RMSE 0 (exact), worst at t=7.27. S2: worst 1.4761, RMSE 0.4475, worst at t=5.00.



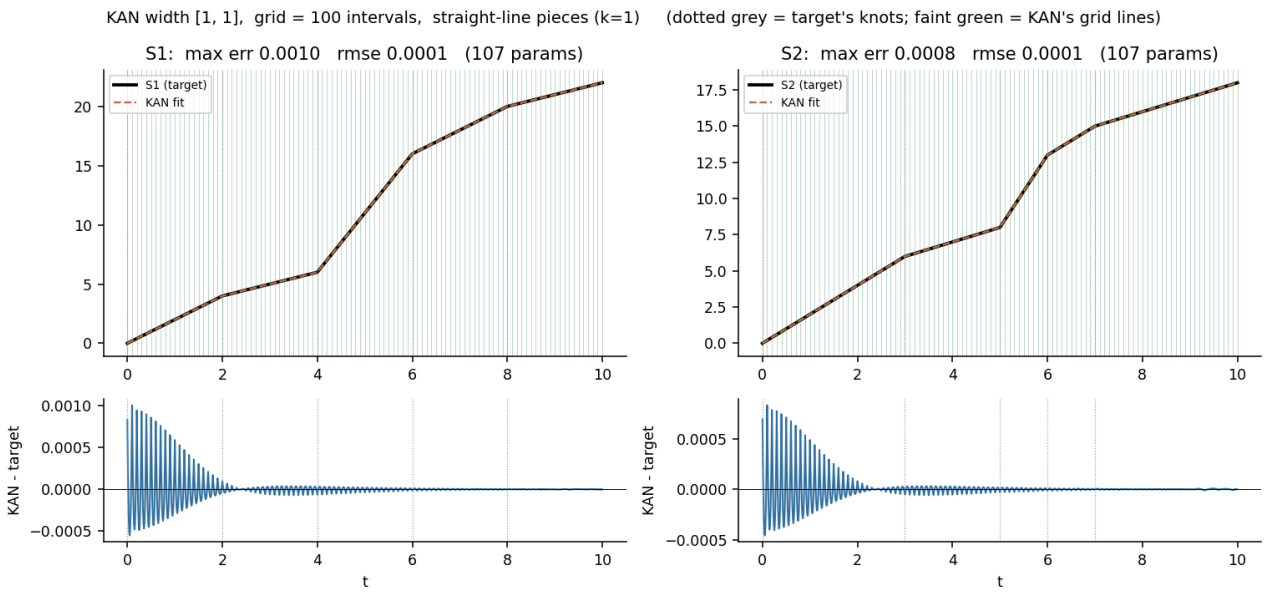
Experiment A. Width [1, 1], grid 10, k=1, 17 parameters. S1: worst 0 (exact), RMSE 0 (exact), worst at t=6.22. S2: worst 0 (exact), RMSE 0 (exact), worst at t=8.81.



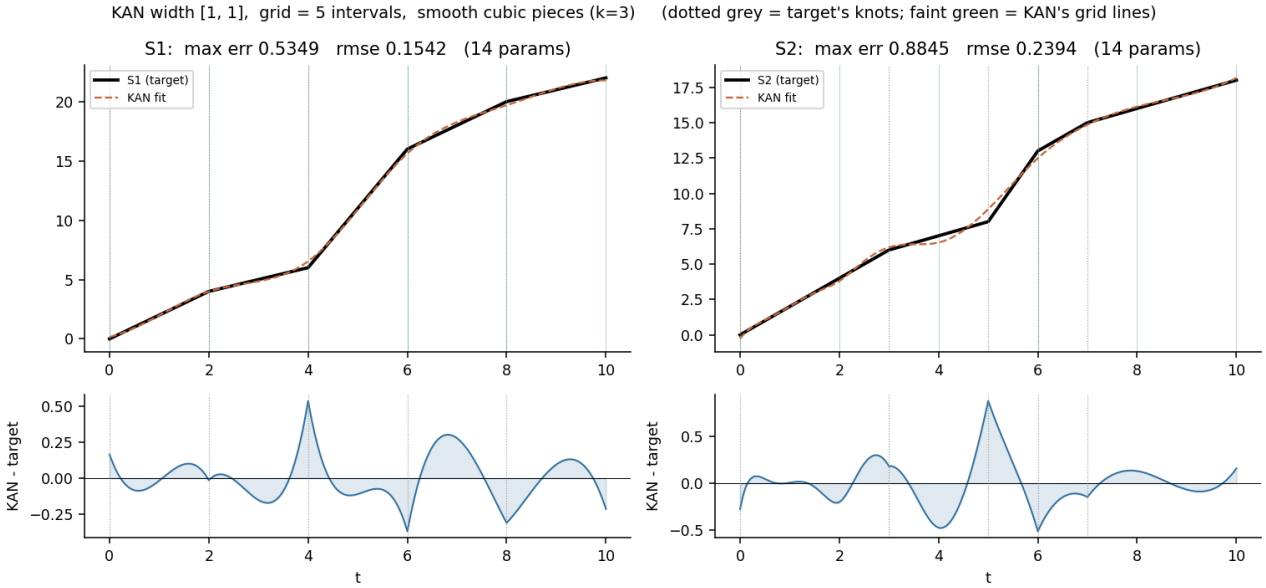
Experiment A. Width [1, 1], grid 20, k=1, 27 parameters. S1: worst 0.0240, RMSE 0.0034, worst at t=0.00. S2: worst 0.0199, RMSE 0.0028, worst at t=0.00.



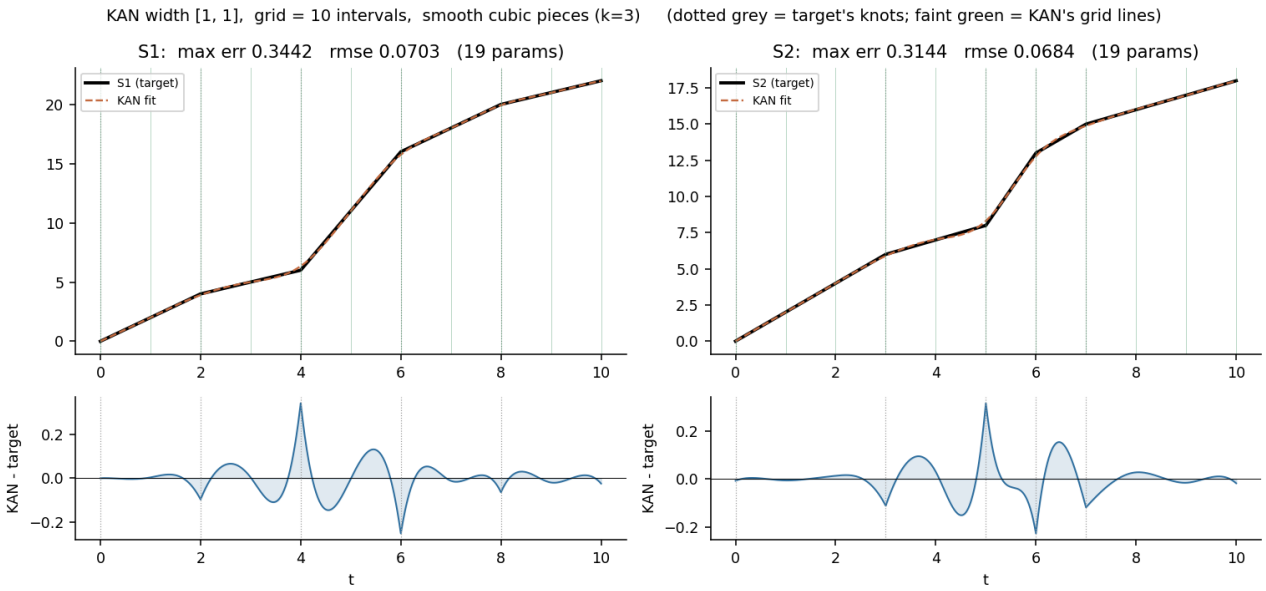
Experiment A. Width [1, 1], grid 50, k=1, 57 parameters. S1: worst 0.0039, RMSE 0.0005, worst at t=0.20. S2: worst 0.0033, RMSE 0.0004, worst at t=0.20.



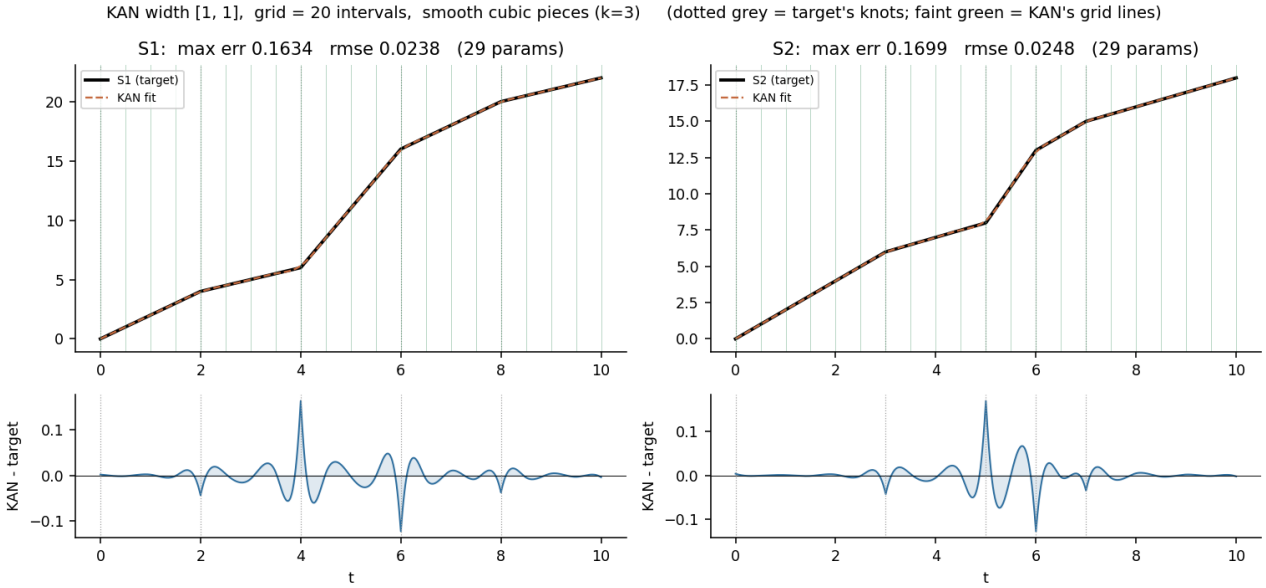
Experiment A. Width [1, 1], grid 100, k=1, 107 parameters. S1: worst 0.0010, RMSE 0.0001, worst at t=0.10. S2: worst 0.0008, RMSE 0.0001, worst at t=0.10.



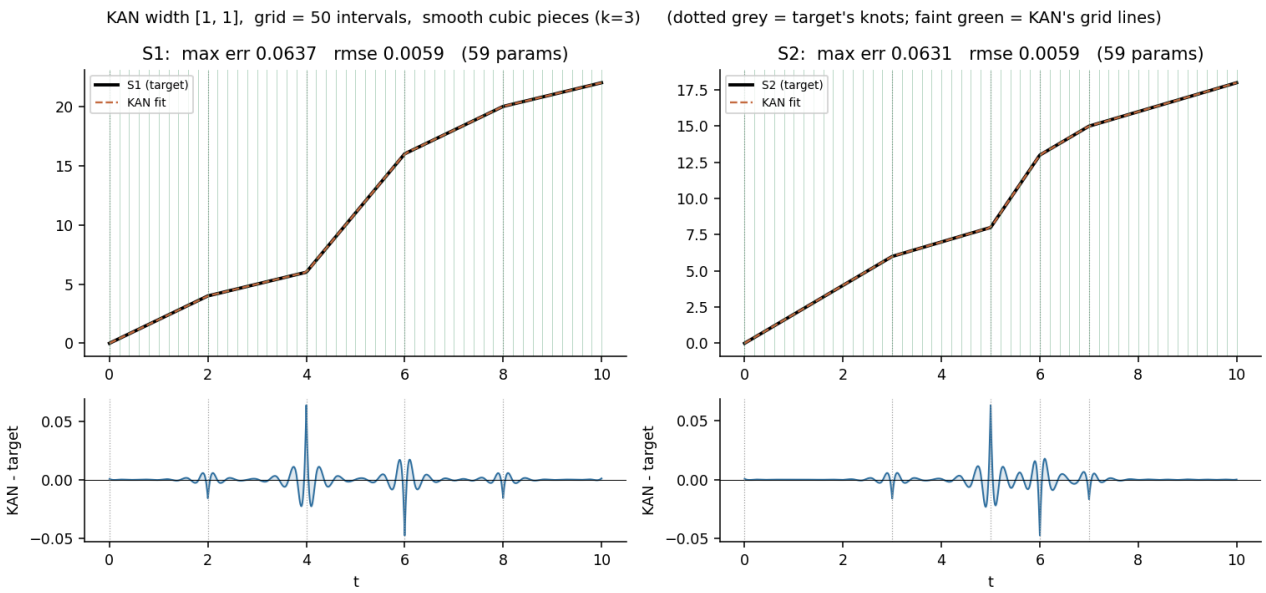
Experiment B. Width [1, 1], grid 5, k=3, 14 parameters. S1: worst 0.5349, RMSE 0.1542, worst at t=4.00. S2: worst 0.8845, RMSE 0.2394, worst at t=5.00.



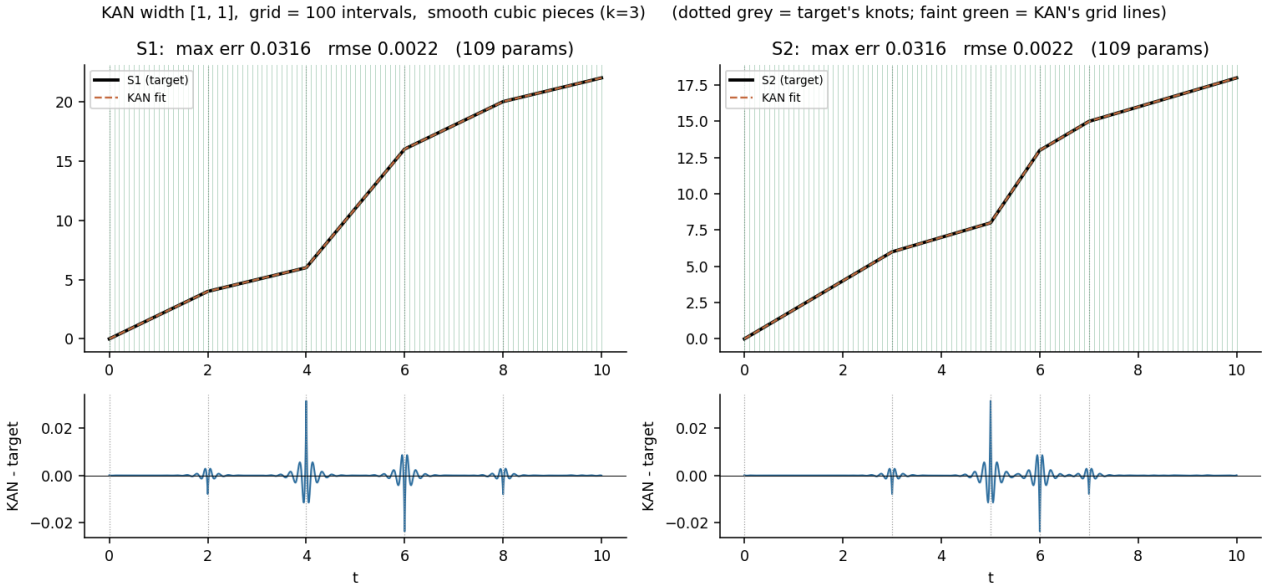
Experiment B. Width [1, 1], grid 10, k=3, 19 parameters. S1: worst 0.3442, RMSE 0.0703, worst at t=4.00. S2: worst 0.3144, RMSE 0.0684, worst at t=5.00.



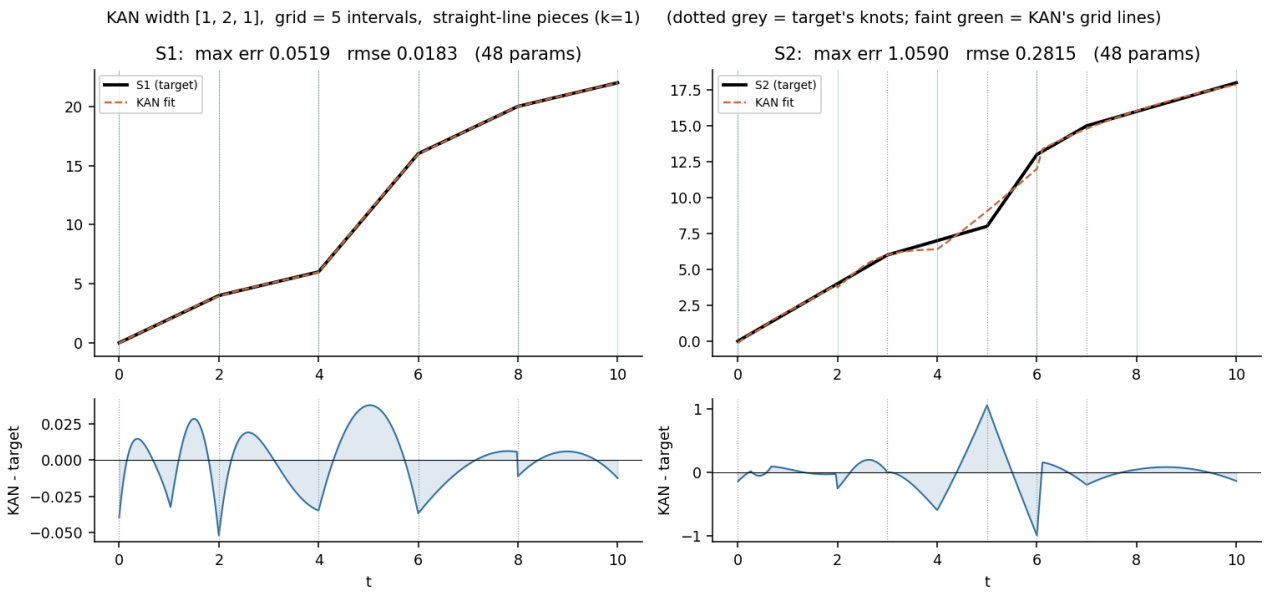
Experiment B. Width [1, 1], grid 20, k=3, 29 parameters. S1: worst 0.1634, RMSE 0.0238, worst at t=4.00. S2: worst 0.1699, RMSE 0.0248, worst at t=5.00.



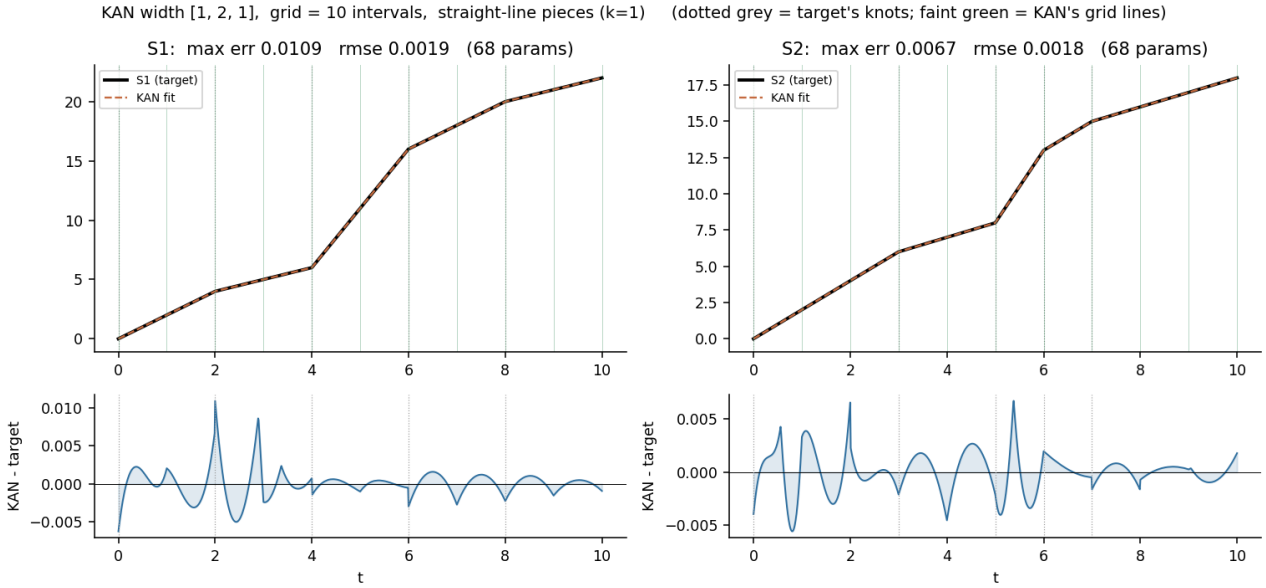
Experiment B. Width [1, 1], grid 50, k=3, 59 parameters. S1: worst 0.0637, RMSE 0.0059, worst at t=4.00. S2: worst 0.0631, RMSE 0.0059, worst at t=5.00.



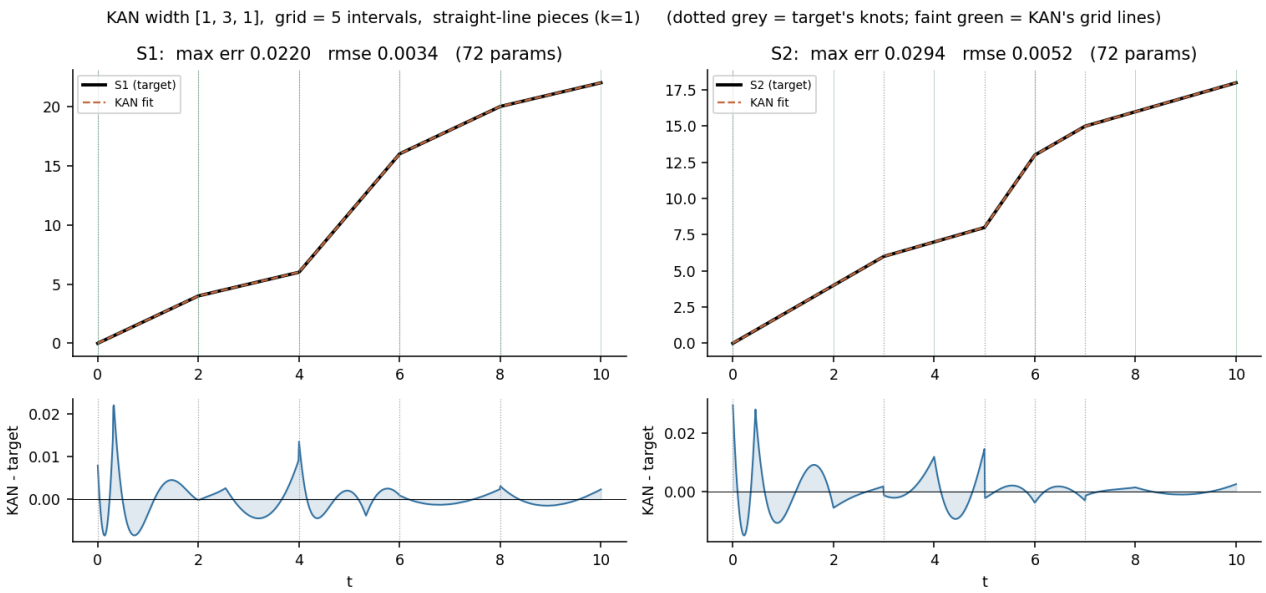
Experiment B. Width [1, 1], grid 100, k=3, 109 parameters. S1: worst 0.0316, RMSE 0.0022, worst at t=4.00. S2: worst 0.0316, RMSE 0.0022, worst at t=5.00.



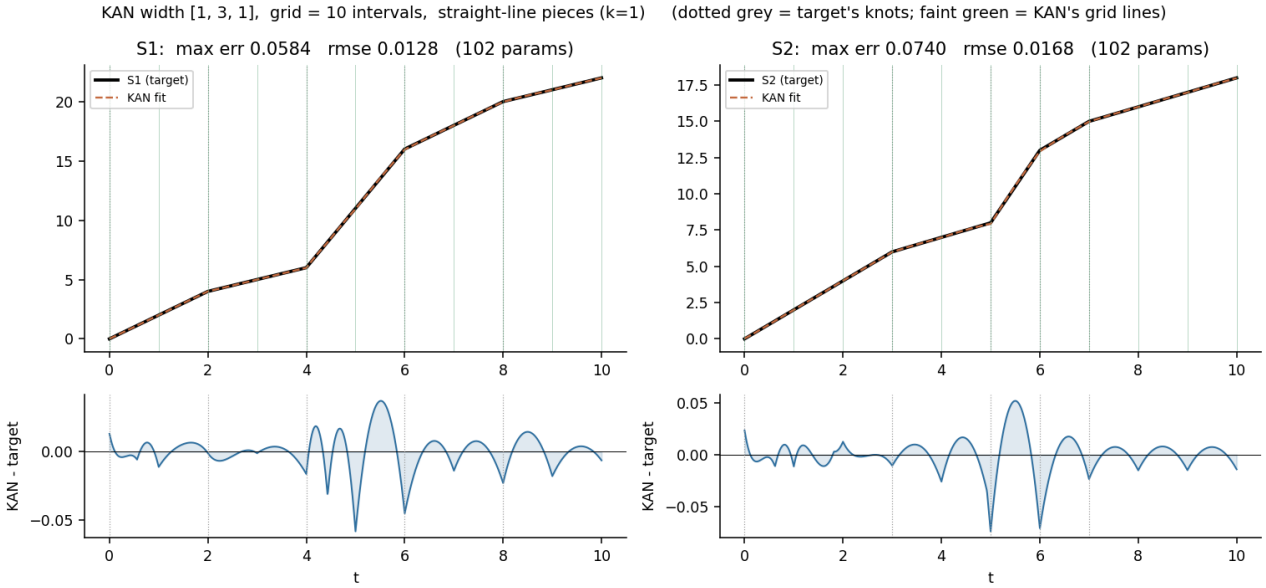
Experiment C. Width [1, 2, 1], grid 5, k=1, 48 parameters. S1: worst 0.0519, RMSE 0.0183, worst at t=2.00. S2: worst 1.0590, RMSE 0.2815, worst at t=5.00.



Experiment C. Width [1, 2, 1], grid 10, k=1, 68 parameters. S1: worst 0.0109, RMSE 0.0019, worst at t=2.00. S2: worst 0.0067, RMSE 0.0018, worst at t=5.38.



Experiment C. Width [1, 3, 1], grid 5, k=1, 72 parameters. S1: worst 0.0220, RMSE 0.0034, worst at t=0.32. S2: worst 0.0294, RMSE 0.0052, worst at t=0.00.



Experiment C. Width [1, 3, 1], grid 10, k=1, 102 parameters. S1: worst 0.0584, RMSE 0.0128, worst at t=5.00. S2: worst 0.0740, RMSE 0.0168, worst at t=5.00.